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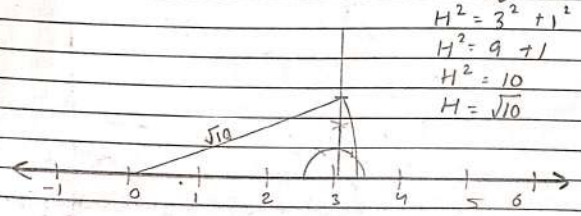
Represent $\sqrt{10}$ on numberline

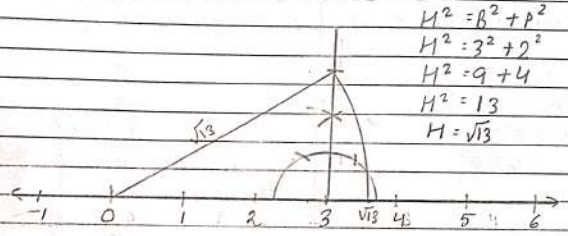
$\sqrt{10}$.

$H^2 = B^2 + P^2$
$H^2 = 3^2 + 1^2$
$H^2 = 9 + 1$
$H^2 = 10$
$H = \sqrt{10}$



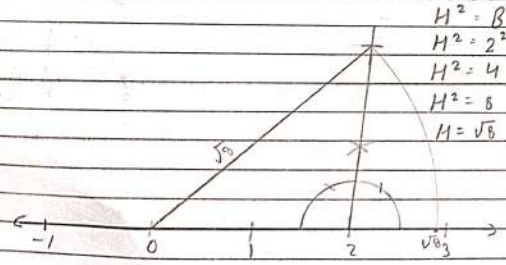
$\sqrt{13}$

$H^2 = B^2 + P^2$
$H^2 = 3^2 + 2^2$
$H^2 = 9 + 4$
$H^2 = 13$
$H = \sqrt{13}$



$\sqrt{8}$

$H^2 = B^2 + P^2$
$H^2 = 2^2 + 2^2$
$H^2 = 4 + 4$
$H^2 = 8$
$H = \sqrt{8}$



Spiral

$\sqrt{68}, \sqrt{20}, \sqrt{17}, \sqrt{7}, \sqrt{37}$

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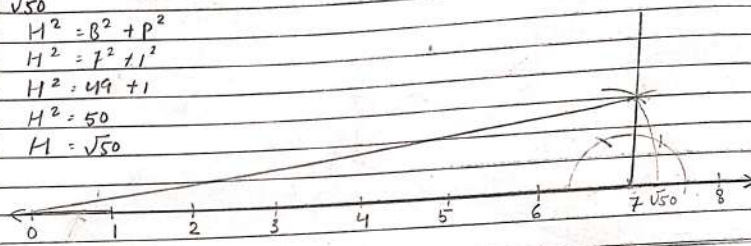
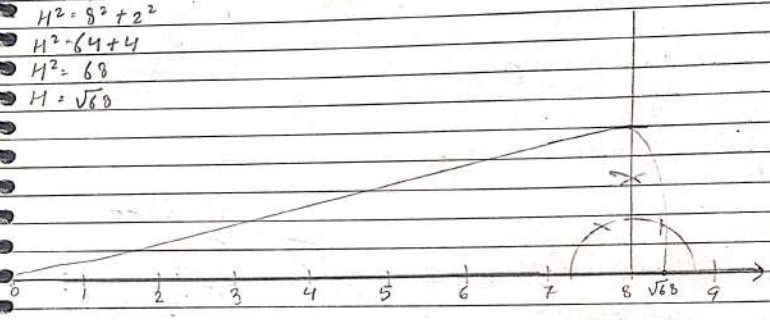
$\sqrt{50}$

$H^2 = B^2 + P^2$
$H^2 = 7^2 + 1^2$
$H^2 = 49 + 1$
$H^2 = 50$
$H = \sqrt{50}$



$\sqrt{68}$

$H^2 = B^2 + P^2$
$H^2 = 8^2 + 2^2$
$H^2 = 64 + 4$
$H^2 = 68$
$H = \sqrt{68}$



Spiral

Date

$\sqrt{20}$ on number line

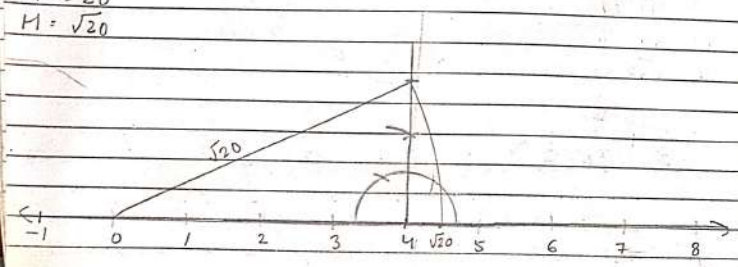
$H^2 = B^2 + P^2$

$H^2 = 4^2 + 2^2$

$H^2 = 16 + 4$

$H^2 = 20$

$H = \sqrt{20}$



$\sqrt{17}$ on number line

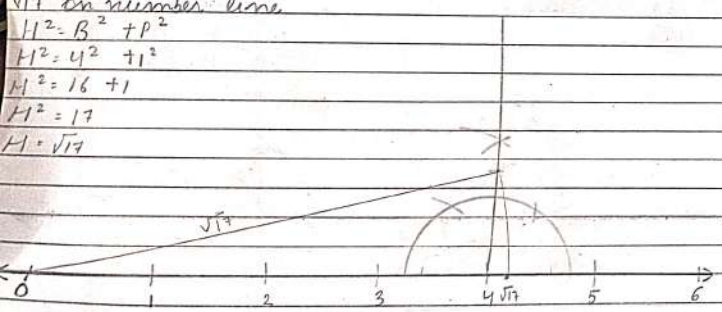
$H^2 = B^2 + P^2$

$H^2 = 4^2 + 1^2$

$H^2 = 16 + 1$

$H^2 = 17$

$H = \sqrt{17}$



Spiral

Date

$\sqrt{7}$

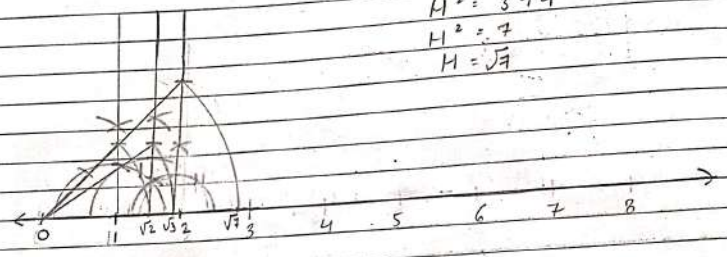
$H^2 = B^2 + P^2$

$H^2 = (\sqrt{3})^2 + (2)^2$

$H^2 = 3 + 4$

$H^2 = 7$

$H = \sqrt{7}$



$\sqrt{37}$

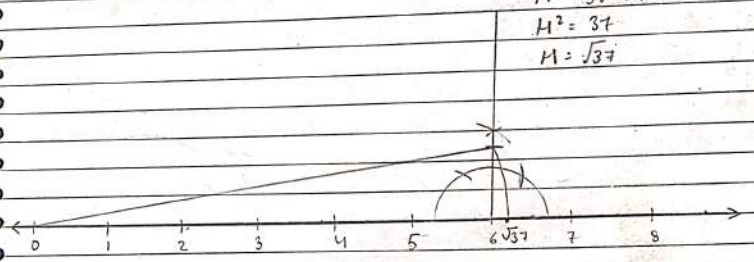
$H^2 = B^2 + P^2$

$H^2 = 6^2 + 1^2$

$H^2 = 36 + 1$

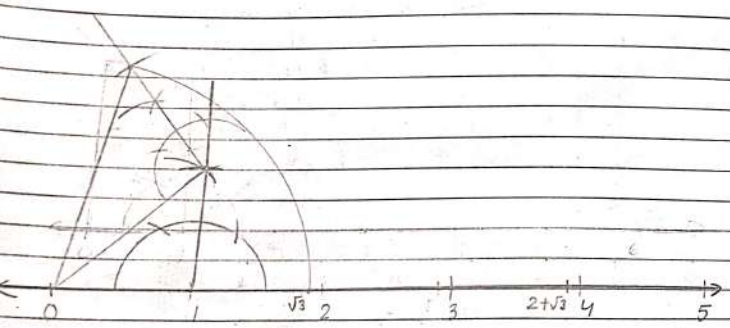
$H^2 = 37$

$H = \sqrt{37}$

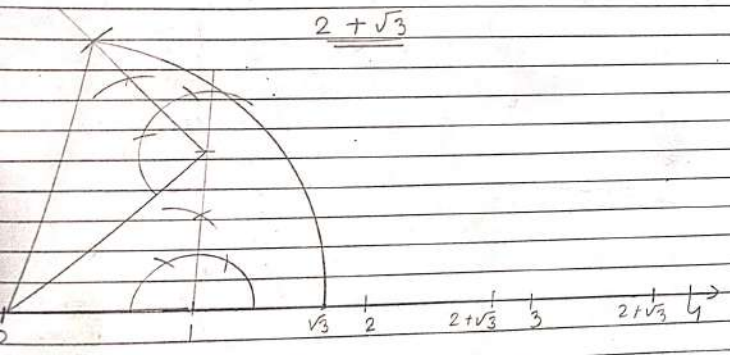


Spiral

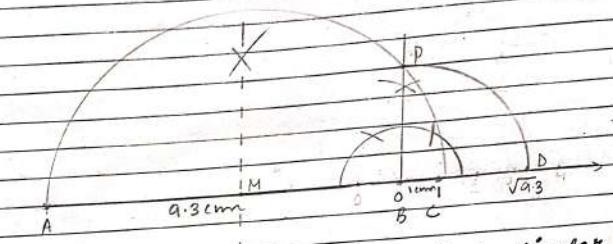
Q. Represent $2+\sqrt{3}$ on no. line.



$2+\sqrt{3}$



Q. Represent $\sqrt{9.3}$ on number line.



- Draw 9.3 cm
- Extend it by 1cm.

- Draw Perpendicular bisector of AC

→ Here, M is the mid-point of AC, Draw a semicircle taking M as centre.

→ Draw right angle at B point

→ taking BP as radius, cut the arc on number line as you can see, it is D point.

- i.e. $BD = \sqrt{9.3}$

Justification

Here, $AC = 10.3$

$AM = CM = \frac{10.3}{2} = 5.15$

$\therefore PM = 5.15$

$BM = CM - 1$

$BM = 5.15 - 1$

$BM = 4.15$

In ΔMBP

$\Rightarrow (MP)^2 = (BM)^2 + (BP)^2$

$\Rightarrow BP^2 = MP^2 - BM^2$

$\Rightarrow BP^2 = (5.15)^2 - (4.15)^2$

$\Rightarrow BP^2 = [5.15 + 4.15][5.15 - 4.15]$

$\Rightarrow BP^2 = [9.3][1] = BP = \sqrt{9.3}$ hence, verified.

Spiral

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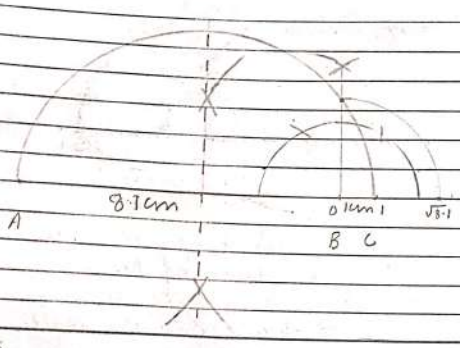
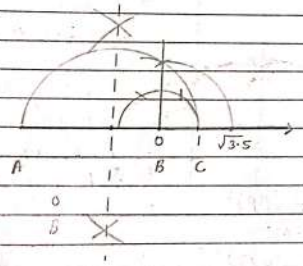
• Represent root 8.1 on number line.



$\sqrt{3.5}$



Spiral

H.W → $\sqrt{4.5}$, $\sqrt{5.6}$, $\sqrt{2.3}$

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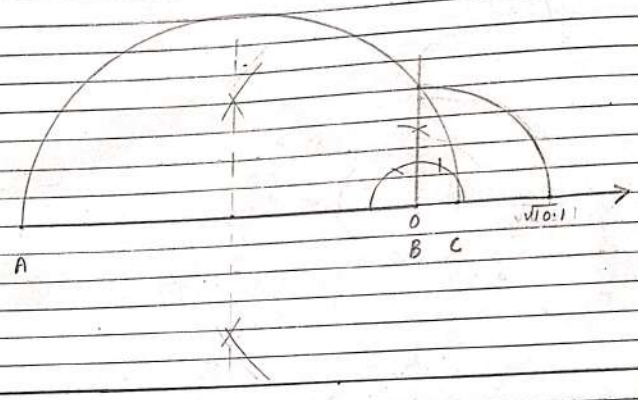
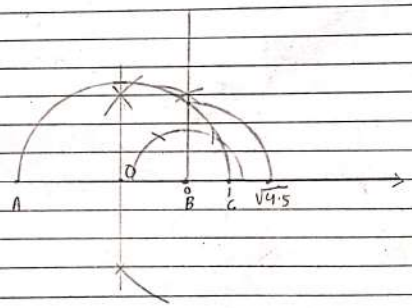
$\sqrt{10.1}$

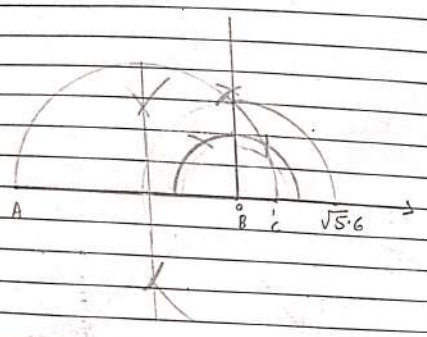


$\sqrt{4.5}$

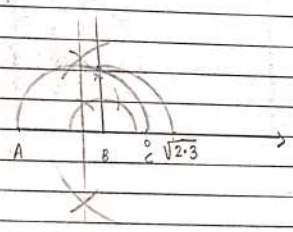


Spiral

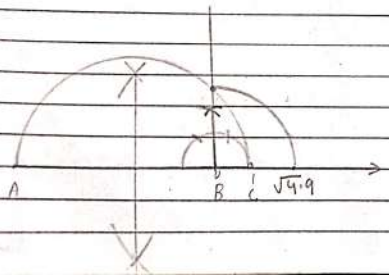
$\sqrt{5.6}$



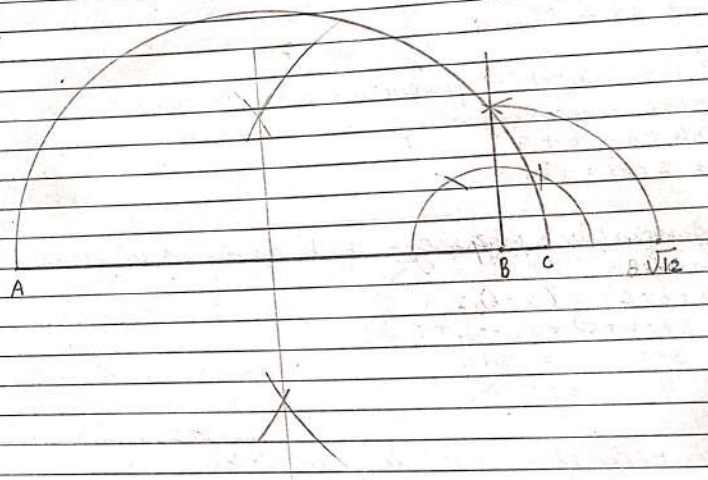
$\sqrt{2.3}$



$\sqrt{4.9}$



$\sqrt{12}$



· Operations on Real numbers.

→ Properties

1. Addition

→ closure property - The sum of two real numbers is always a real no.

$a + b = c$ → Real no.

↓ Real no.

Real no.

Eg: → $3 + 2 = 5$

2 → Commutative property - here a and b are real numbers.

→ $a + b = b + a$

Eg: → $2 + \sqrt{3} = \sqrt{3} + 2$

→ Associative property - a, b, and c are real numbers

→ $a + (b + c) = (a + b) + c$

Eg: → $3 + (2 + 5) = (3 + 2) + 5$

$3 + 7 = 5 + 5$

$10 = 10$

→

→ Identity element for addition

a is Real no.

⇒ $a + 0 = a$

Eg: $5 + 0 = 5$

↓

Identity element for addition

→ Additive inverse

if 'a' is Real no.

⇒ $a + (-a) = 0$

then, $-a$ is the additive inverse.

Eg: $5 + (-5) = 0$

2- Multiplication:

→ closure property

$a \times b = c$ → Real no.

↓ Real no.

Real no.

Eg → $3 \times \sqrt{2} = 3\sqrt{2}$

→ commutative property

$a \times b = b \times a$

Eg → $2 \times \sqrt{5} = \sqrt{5} \times 2$

→ Associative property

$a \times (b \times c) = (a \times b) \times c$

Eg → $2 \times (3 \times 4) = (2 \times 3) \times 4$

$24 = 24$

→ Multiplicative identity.

$a \times 1 = a$

Eg → $4 \times 1 = 4$